



起升机构设计计算书 Calculation Sheet for Design of Hoisting Mechanism

1.起升机构

1. Hoisting Mechanism

机构组别: M8

Mechanism group: M8

T7 L4

额定起重量: $LL=10.2t$

Rated hoisting load: $LL=10.2t$

吊具及上架重量: $LS=7.8t$

Spreader and headblock weight: $LS=7.8t$

钢丝绳下起吊重量: $Q=18t$

Hoisting load under wire rope: $Q=18t$

起升载荷: $P_Q=176.58kN$

Hoisting load: $P_Q=176.58kN$

额定起升速度: $V_{q1}=50m/min$

Rated hoisting speed: $V_{q1}=50m/min$

$=0.833m/s$

空载起升速度: $V_{q2}=50m/min$

No-load hoisting speed: $V_{q2}=50m/min$

$=0.83m/s$

最大起升高度: $H_{max}=30000mm$

Maximum hoisting height: $H_{max}=30,000mm$

允许的起动时间: $[t_z]=3.0s$ (满载)

Allowable starting time: $[t_z]=3.0s$ (full load)

$[t_q]=3.0s$ (空载)

$[t_q]=3.0s$ (no-load)

1.1.起升钢丝绳

1.1 Hoisting Wire Rope

采用滚动轴承的滑轮效率: $\eta_1=0.98$

Pulley efficiency using the rolling bearing: $\eta_1=0.98$

导向滑轮的效率: $\eta_D=1$

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Efficiency of the guide pulley: $\eta_D=1$

钢丝绳传动效率: $\eta_1=1.00 \left(\eta_1 = \frac{1-\eta_l^a}{1-\eta_l} \cdot \eta_D \right)$

Transmission efficiency of wire rope: $\eta_1=1.00 \left(\eta_1 = \frac{1-\eta_l^a}{1-\eta_l} \cdot \eta_D \right)$

起升钢丝绳安全系数: $n=9$ (机构级别 M8)

Safety coefficient of hoisting wire rope: $n=9$ (mechanism class: M8)

钢丝绳倍率: $a=1$

Multiple ratio of wire rope: $a=1$

钢丝绳的分支数: $m=4$

The number of branches of the wire rope: $m=4$

钢丝绳与铅锤面的夹角: $\alpha_1=0^\circ$

Included angle between wire rope and plumb face: $\alpha_1=0^\circ$

单根钢丝绳拉力: $S = 44.145\text{kN}$

$$S = \frac{P_Q}{m\eta_1}$$
$$S = \frac{P_Q}{m\eta_1}$$

Tension of single wire rope: $S = 44.145\text{kN}$

所选的钢丝绳直径 (mm): $d=20\text{mm}$

Diameter of selected wire rope (mm): $d=20\text{mm}$

最小破断拉力 (kN): $S_{\max}=397.31\text{kN}$

Minimum breaking force (kN): $S_{\max}=397.31\text{kN}$

钢丝绳公称破断拉力 (kN): $[S]=442.2\text{kN}$ 钢丝绳满足使用要求!

Nominal breaking force of wire rope (kN): $[S]=442.2\text{kN}$ The wire rope meets the use requirements!

钢丝绳型号: VEROTECH 10- ϕ 20-2160

Wire rope model: VEROTECH 10- ϕ 20-2160

1.2.卷筒

1.2 Drum

1.2.1.卷筒设计

1.2.1 Drum Design

筒绳直径比系数: $h=25$ (机构级别 M8), 两个双联卷筒 (四绳抓斗)

Diameter ratio coefficient of drum rope: $h=25$ (mechanism class: M8), two dual-sensing drums (four-rope grab)

卷筒的最小直径 (mm): $D_{\min}=500$

Minimum diameter of the drum (mm): $D_{\min}=500$

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所选卷筒的名义直径 (mm): $D_0 = 520\text{mm}$ (考虑卷筒联轴器的安装)

Nominal diameter of the selected drum (mm): $D_0 = 520\text{mm}$ (considering the installation of drum coupling)

卷筒的槽底径: $D_1 = 500\text{mm}$

Slot base diameter of drum: $D_1 = 500\text{mm}$

卷筒内径: $D_2 = 468\text{mm}$

Inner diameter of drum: $D_2 = 468\text{mm}$

标准槽时的卷筒槽深 (mm): $H_1 = 10\text{mm}$ $H_1 = (0.25 \sim 0.4)d$

Depth of the drum slot in the case of the standard slot (mm): $H_1 = 10\text{mm}$
 $H_1 = (0.25 \sim 0.4)d$

标准槽卷筒的节距 (mm): $p = 26\text{mm}$ $p \approx d + (2 \sim 4)$

Pitch of standard slot drum (mm): $p = 26\text{mm}$ $p \approx d + (2 \sim 4)$

预留安全圈数: $n_s = 3$ (3 圈安全圈)

The number of reserved safety grooves: $n_s = 3$ (three safety grooves)

卷筒切削螺旋部分长度: $L_0 = 529\text{mm}$

Length of the cutting spiral part of the drum: $L_0 = 529\text{mm}$

起升圈数 18 圈

The number of hoisting grooves: 18 Pcs.

标准槽压板尺寸: $L_1 = 52\text{mm}$

Clamp plate size of standard slot: $L_1 = 52\text{mm}$

安全圈钢丝绳所需长度: $L_2 = 78\text{mm}$ $L_2 = 3 \times p$

Length of wire rope required for safety grooves: $L_2 = 78\text{mm}$ $L_2 = 3 \times p$

卷筒中间光滑部分长度: $L_g = 820\text{mm}$

Length of the smooth part in the middle of the drum: $L_g = 820\text{mm}$

单层卷绕卷筒的理论长度 (双联): $L = 2L_0 + 2L_1 + 2L_2 + L_g$

Theoretical length of single-layer winding drum (dual-sensing):

$$L = 2L_0 + 2L_1 + 2L_2 + L_g$$

$$= 2191\text{mm}$$

实际选用的卷筒长度: $[L] = 2200\text{mm}$

Length of the actual selected drum: $[L] = 2200\text{mm}$

卷筒壁厚: $\delta \approx d = 16\text{mm}$

Drum wall thickness: $\delta \approx d = 16\text{mm}$

卷筒重量: $m_G \approx 940.00\text{kg}$

Drum weight: $m_G \approx 940.00\text{kg}$

卷筒转动惯量: $J_G = 230.03\text{kg}\cdot\text{m}$

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Rotational inertia of drum: $J_G=230.03\text{kg.m}$

1.2.2.卷筒强度校核

1.2.2 Drum Strength Check

$L/D=4.23 > 3$

系数 $A_1=1$ (单层卷绕)

Coefficient: $A_1=1$ (single-layer winding)

最大表面压应力(MPa): $\sigma_c=79.59 < 230$

Maximum surface compressive stress (MPa): $\sigma_c=79.59 < 230$

$L/D=4.23 > 3$

卷筒最大弯矩: $M=1.10\text{E}+05 \text{ N.m}$

Maximum bending moment of drum: $M=1.10\text{E}+05 \text{ N.m}$

卷筒最大扭矩: $T=8.26\text{E}+04 \text{ N.m}$

Maximum torque of drum: $T=8.26\text{E}+04 \text{ N.m}$

卷筒截面抗弯模量: $W=2.91\text{E}-03 \text{ m}^3$

Section bending modulus of drum: $W=2.91\text{E}-03 \text{ m}^3$

折算应力: $\sigma_p=47.21 < 172.5 \text{ MPa}$ 满足要求!

Reduced stress: $\sigma_p=47.21 < 172.5 \text{ MPa}$ Meet the requirements!

1.2.3.卷筒稳定性校核

1.2.3 Drum Stability Check

$D \geq 1200\text{mm}$, 且 $L > 2D$ 时, 须对卷筒壁进行稳定性验算

In case of $D \geq 1200\text{mm}$ and $L > 2D$, the stability of the drum must be checked.

卷筒壁单位压力: $p=6.79\text{MPa}$

Unit pressure drum wall: $p=6.79\text{MPa}$

临界单位压力: $p_{cr}=13.76\text{MPa}$ (钢卷筒)

Critical unit pressure: $p_{cr}=13.76\text{MPa}$ (steel drum)

稳定性系数: $K=2.03 \geq 1.3 \sim 1.5$ $K=\frac{P_{cr}}{P}=1.3 \sim 1.5$ 满足要求!

Stability coefficient: $K=2.03 \geq 1.3 \sim 1.5$ $K=\frac{P_{cr}}{P}=1.3 \sim 1.5$ Meet the requirements!

1.2.4.卷筒轴的设计

1.2.4 Design of Drum Shaft

机构增大系数: $\gamma_m=1.30$ (M8)

Mechanism increase coefficient: $\gamma_m=1.30$ (M8)

卷筒轴上作用的支承反力: 水平分量=0.00 kN

Supporting reaction force effecting on the drum shaft: horizontal component =0.00kN

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垂直分量=64.00kN

Vertical component =64.00kN

$R_A=64.00\text{kN}$

支承距离: $L_2=110.0\text{ mm}$

Support distance: $L_2=110.0\text{mm}$

$L_3=218.0\text{ mm}$

支承处的直径 $d=200.0\text{ mm}$

Diameter of supporting part: $d=200.0\text{mm}$

轴惯性矩 $I=7.85\text{E}+07\text{ mm}^4$

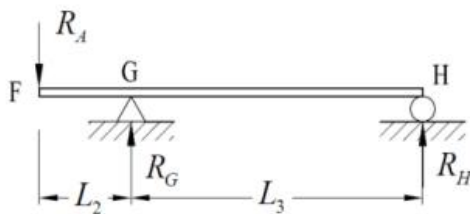
Axial moment of inertia: $I=7.85\text{E}+07\text{ mm}^4$

最大弯矩 $M=7040\text{ kN}\cdot\text{mm}$

Maximum bending moment: $M=7,040\text{ kN}\cdot\text{mm}$

最大应力 $\sigma=8.97\text{ MPa}$ 满足要求!

Maximum stress: $\sigma=8.97\text{ MPa}$ Meet the requirements!



(c) 卷筒轴 Drum shaft

1.2.5.卷筒的转速

1.2.5 Drum Speed

卷筒的理论转速: $n_j = \frac{a \cdot v_q}{\pi D_0}$

Theoretical speed of the drum: $n_j = \frac{a \cdot v_q}{\pi D_0}$

满载时: $n_j=30.61\text{r/min}$

Full load: $n_j=30.61\text{r/min}$

空载时: $n_j=30.61\text{r/min}$

No-load: $n_j=30.61\text{r/min}$

1.3.电动机

1.3 Motor

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1.3.1.选型

1.3.1 Model Selection

电动机台数: $M=1.0$

Number of motors: $M=1.0$

卷筒效率: $h_2=0.96$

Drum efficiency: $h_2=0.96$

减速器传动效率: $h_2=0.96$

Reducer transmission efficiency: $h_2=0.96$

起升机构总效率: $h_z=0.922$

Overall efficiency of hoisting mechanism: $h_z=0.922$

起升机构的静功率: $P_N = \frac{P_Q \cdot v_q}{1000\eta} = 159.67 \text{ kW}$

Static power of hoisting mechanism: $P_N = \frac{P_Q \cdot v_q}{1000\eta} = 159.67 \text{ kW}$

所选电动机的参数如下:

The parameters of selected motor are as follows:

型号: **YZPF315M1-4**

Model: **YZPF315M1-4**

额定功率: $P_n=160\text{kW}$

Rated power: $P_n=160\text{kW}$

额定转速: $n_n=1488\text{r/min}$

Rated speed: $n_n=1488\text{r/min}$

额定转矩: $M_n=1026.2\text{N.m}$

Rated torque: $M_n=1026.2\text{N.m}$

电动机最大转矩倍数: $\lambda_m=2.6$

Maximum torque multiple of the motor: $\lambda_m=2.6$

峰值扭矩: $M_A=2668.1\text{N.m}$

Peak torque: $M_A=2668.1\text{N.m}$

电动机输出轴轴径: $d_{d1}=80\text{mm}$

Diameter of the motor output shaft: $d_{d1}=80\text{mm}$

输出轴轴端长度: $l_{d1}=170\text{mm}$

Output shaft end length: $l_{d1}=170\text{mm}$

电动机效率: $\eta_d=0.947$

Motor efficiency: $\eta_d=0.947$

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转动惯量: $J_1=4.02\text{kg.m}^2$

Rotational inertia: $J_1=4.02\text{kg.m}^2$

电动机重量: $m_1=1040\text{kg}$

Motor weight: $m_1=1040\text{kg}$

所选电机满足要求!

The selected motor meets the requirements!

1.3.2.过载校核

1.3.2 Overload Check

$$P_n \geq \frac{H}{m\lambda_m} \cdot \frac{P_Q v_q}{1000\eta}$$

系数: $H=2.2$

Coefficient

$$\frac{H}{m\lambda_m} \cdot \frac{P_Q v_q}{1000\eta} = 135.10 \quad \text{kW}$$

满足要求!

Meet the requirements!

1.3.3.发热校核

1.3.3 Heating Check

$$P_s \geq G \frac{P_Q v_q}{1000\eta}$$

功率修正系数: $K=0.9$ (环境温度 50°C)

Power correction coefficient: $K=0.9$ (ambient temperature: 50°C)

稳态负载系数: $G=0.8$ (G2)

Steady-state load coefficient: $G=0.8$ (G2)

$$G \frac{P_Q v_q}{1000\eta} = 127.73 \quad \text{kW}$$

满足要求!

$$G \frac{P_Q v_q}{1000\eta} = 127.73 \quad \text{kW}$$

Meet the requirements!

1.4.减速器

1.4 Reducer

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1.4.1. 减速器选型

1.4.1 Reducer Selection

起升机构的计算传动比: $i_0=48.62$ ($i_0=n_n/n_{\text{卷}}$)

Calculation transmission ratio of hoisting mechanism: $i_0=48.62$ ($i_0=n_n/n_{\text{roll}}$)

服务系数: $f_B=1.10$

Service coefficient: $f_B=1.10$

原动机系数: $f_M=1.00$ (电动机驱动)

Prime motor: $f_M=1.00$ (motor drive)

尖峰负载系数: $f_s=0.70$ (单向, 每小时峰值负载次数 21-40)

Peak load coefficient: $f_s=0.70$ (unidirectional, number of peak loads per hour: 21-40)

环境系数: $f_4=1.00$

Environmental coefficient: $f_4=1.00$

海拔系数: $f_H=1.00$ (海拔 1000m 以下)

Altitude coefficient: $f_H=1.00$ (altitude: 1000m and below)

工作制系数: $f_{ED}=1.19$

Working system coefficient: $f_{ED}=1.19$

减速器的额定功率应满足: $P_{N1}=P_n \times f_B \times f_M = 88.0 \quad \text{kW}$

The rated power of the reducer shall meet: $P_{N1}=P_n \times f_B \times f_M = 88.0 \quad \text{kW}$

减速器的输入转矩应满足: $T_{N1}=T_n \times f_B \times f_M = 564.4 \quad \text{N}\cdot\text{m}$

The input torque of the reducer shall meet: $T_{N1}=T_n \times f_B \times f_M = 564.4 \quad \text{N}\cdot\text{m}$

减速器的输出转矩应满足: $T_{N2}=9.55 \frac{P_n}{n_j} \times f_B \times f_M = 27.5 \quad \text{kN}\cdot\text{m}$

The output torque of the reducer shall meet: $T_{N2}=9.55 \frac{P_n}{n_j} \times f_B \times f_M = 27.5 \quad \text{kN}\cdot\text{m}$

所选减速器的参数如下:

The parameters of the selected reducer are as follows:

型号: **SK9321EV**

Model: **SK9321EV**

台数: $m_j=2$

Qty.: $m_j=2$

传动比: $i=49.13^*$

Transmission ratio: $i=49.13^*$

许用转速: $[n_n]=1500\text{r/min}$

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Allowable rotating speed: $[n_n]=1500\text{r/min}$

高速轴许用功率: $[P_{N1}]=132.00\text{kW}$

Allowable power of high-speed shaft: $[P_{N1}]=132.00\text{kW}$

热额定功率: $[P_{G1}]=106.00\text{kW}$ (1500r/min, 40°C, 不带辅助冷却装置)

Rated thermal power: $[P_{G1}]=106.00\text{kW}$ (1500r/min, 40°C, without auxiliary cooling device)

输出轴许用转矩: $[M_{2\max}]=41.35\text{kN.m}$

Allowable torque of the output shaft: $[M_{2\max}]=41.35\text{kN.m}$

作用力系数: $K=1$ (支承距离 $x=195$)

Effecting force coefficient: $K=1$ (support distance: $x=195$)

输出轴上允许径向载荷: $[F_R]=80\text{kN}$

Allowable radial load on the output shaft: $[F_R]=80\text{kN}$

允许夹角: 35°

Allowable included angle: 35°

机座号: SX9.21

Seat No.: SX9.21

卷筒和电动机轴中心距: $a=718\text{mm}$

Center distance between the drum and the motor shaft: $a=718\text{mm}$

输入轴直径: $d_{12}=55\text{mm}$

Input shaft diameter: $d_{12}=55\text{mm}$

输入轴轴端长度: $l_{12}=110\text{mm}$

Input shaft end length: $l_{12}=110\text{mm}$

输出轴直径: $d_{10}=160\text{mm}$

Output shaft diameter: $d_{10}=160\text{mm}$

输出轴轴端长度: $l_{10}=300\text{mm}$

Output shaft end length: $l_{10}=300\text{mm}$

转动惯量: $J_2=0.044\text{kg.m}^2$

Rotational inertia: $J_2=0.044\text{kg.m}^2$

1.4.2.峰值负载校核

1.4.2 Peak Load Check

峰值扭矩: $T_{\max} f_s \leq [M_{2\max}]$

Peak torque: $T_{\max} f_s \leq [M_{2\max}]$

$T_{\max} f_s = 22.2\text{kN.m}$ 满足要求!

$T_{\max} f_s = 22.2\text{kN.m}$ Meet the requirements!

1.4.3.热功率校核

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1.4.3 Thermal Power Check

许用热功率: $P_{wg} \leq [P_{G1}] \times f_4 = 106 \text{ kW}$

Allowable thermal power: $P_{wg} \leq [P_{G1}] \times f_4 = 106 \text{ kW}$

$$P_{wg} = P_n \times f_v \times f_H \times f_{ED} = 95.2 \text{ kW} \quad \text{满足要求!}$$

$$P_{wg} = P_n \times f_v \times f_H \times f_{ED} = 95.2 \text{ kW} \quad \text{Meet the requirements!}$$

1.4.4. 输出轴径向力校核

1.4.4 Check for the Radial Force of the Output Shaft

起升状态级别: HC2

Level of hoisting state: HC2

起升动载系数的最小值: $\Phi_{2\min} = 1.10$

Minimum hoisting dynamic load coefficient: $\Phi_{2\min} = 1.10$

由起升状态级别设定的系数: $\beta_2 = 0.34$

Coefficient set by the level of hoisting state: $\beta_2 = 0.34$

起升载荷动载系数: $\Phi_2 = 1.383$

Dynamic load coefficient of hoisting load: $\Phi_2 = 1.383$

卷筒在减速器轴上的支承距离: $x = 90.000 \text{ mm}$

Support distance of drum on the reducer shaft: $x = 90.000 \text{ mm}$

作用力系数: $K = 1.1$

Effecting force coefficient: $K = 1.1$

低速轴允许径向载荷: $[F_R] = 88 \text{ kN}$

Allowable radial load of low-speed shaft: $[F_R] = 88 \text{ kN}$

作用在减速器轴上的径向载荷: $F_{\max} = R_A = 64 \text{ kN}$

Radial load effecting on the reducer shaft: $F_{\max} = R_A = 64 \text{ kN}$

作用在减速器轴上的转矩: $T_{\max} = \phi_2 \cdot S \cdot D_0 = 31.8 \text{ kN} \cdot \text{m}$

Torque effecting on the reducer shaft: $T_{\max} = \phi_2 \cdot S \cdot D_0 = 31.8 \text{ kN} \cdot \text{m}$

减速机选择满足要求!

The reducer selection meets the requirements!

1.4.5. 实际电动机转速

1.4.5 Actual Motor Speed

转速校核公式:
$$n = \frac{a \cdot v_q}{\pi D_0} \cdot i$$

Speed check formula:
$$n = \frac{a \cdot v_q}{\pi D_0} \cdot i$$

满载电动机转速: $n_{S1} = 1504 \text{ r/min}$

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Position: Mechanical Engineer



Full load motor speed: $n_{S1}=1,504\text{r/min}$

空载电动机转速: $n_{S2}=1504\text{r/min}$

No-load motor speed: $n_{S2}=1,504\text{r/min}$

1.5.联轴器

1.5 Coupling

1.5.1.高速轴联轴器

1.5.1 High-speed Coupling

$$M_c = n \cdot M_{I_{\max}} = n \cdot \Phi_8 \cdot M_n$$

式中: M_n ——电动机额定转矩传到联轴器上的转矩。

In the formula: M_n - rated torque of the motor, torque transmitted to the coupling.

安全系数: $n=1.5$ (起升机构 M8)

Safety coefficient: $n=1.5$ (hoisting mechanism: M8)

刚性动载系数: $\Phi_8=2.0$

Rigid dynamic load factor: $\Phi_8=2.0$

联轴器需传递的转矩: $M_c=3078.6\text{ N.m}$

Torque to be transmitted by the coupling: $M_c=3078.6\text{ N.m}$

所选联轴器的参数如下:

The parameters of the selected coupling are as follows:

型号: **KTR 联轴器 NANA3**

Model: **KTR coupling NANA3**

许用转速: $[n_t]=1900\text{ r/min}$

Allowable rotating speed: $[n_t]=1900\text{ r/min}$

联轴器许用转矩: $[M_t]=4600\text{ N.m}$

Allowable torque of coupling: $[M_t]=4600\text{ N.m}$

轴孔内径: d_1 =电动机 80mm 减速机 55mm

Inner diameter of shaft hole: $d_1=80\text{mm}$ for motor, 55mm for reducer

轴孔长度: l_1 =电动机 117mm 减速机 112mm

Length of shaft hole: $l_1=117\text{mm}$ for motor, 112mm for reducer

转动惯量: $J_{21}=1.24\text{ kg.m}^2$

Rotational inertia: $J_{21}=1.24\text{ kg.m}^2$

重量: $m_{21}=\text{kg}$

Weight: $m_{21}=\text{kg}$

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个数: $n_{21}=2$ 个

Qty.: $n_{21}=2$ Pcs.

联轴器选择满足要求!

The coupling selection meets the requirements!

1.5.2.低速轴联轴器

1.5.2 Low-speed Shaft Coupling

$$M_c = K_2 \cdot M_{\max} = K_2 \cdot \phi_6 \cdot \frac{P_t D_0}{2a\eta_1}$$

式中: P_t ——试验载荷, 为 1.1 倍的起升载荷。 $P_t = 1.1P_Q$

In the formula, P_t - test load, 1.1 times of hoisting load. $P_t = 1.1P_Q$

径向载荷补偿系数: $K_1=2.6$

Compensation coefficient of radial load: $K_1=2.6$

工作级别系数: $K_2=1$

Working class coefficient: $K_2=1$

动载系数: $\Phi_6=1.38$

Dynamic load coefficient: $\Phi_6=1.38$

联轴器需传递的转矩: $M_c=34.8\text{kN.m}$

Torque to be transmitted by the coupling: $M_c=34.8\text{kN.m}$

联接处允许承受的径向载荷: $[F_r]=226\text{kN}$

Allowable radial load at the joint: $[F_r]=226\text{kN}$

所选联轴器的参数如下:

The parameters of the selected coupling are as follows:

型号: **WZL08F**

Model: **WZL08F**

许用转速: $[n_t]=200\text{r/min}$

Allowable rotating speed: $[n_t]=200\text{r/min}$

联轴器许用转矩: $[M_t]=35.5\text{kN.m}$

Allowable torque of coupling: $[M_t]=35.5\text{kN.m}$

联轴器最大许用径向力: $[F_r]=224\text{kN}$

Maximum allowable radial force of coupling: $[F_r]=224\text{kN}$

孔直径: $d_c=300\text{mm}$

Hole diameter: $d_c=300\text{mm}$

孔长度: $l_c=345\text{mm}$

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Hole length: $lc=345\text{mm}$

转动惯量: $J_{22}=2.66\text{kg}\cdot\text{m}^2$

Rotational inertia: $J_{22}=2.66\text{kg}\cdot\text{m}^2$

重量: $m_{22}=134\text{kg}$

Weight: $m_{22}=134\text{kg}$

个数: $n_{22}=2$ 个

Qty.: $n_{22}=2$ Pcs.

联轴器选择满足要求!

The coupling selection meets the requirements!

1.6.制动器

1.6 Brake

1.6.1.高速轴制动器

1.6.1 High-speed Axle Brake

$$M_Z = K_Z \frac{P_Q \cdot D_0 \cdot \eta}{2m_z a \cdot i}$$

制动安全系数: $K_Z=1.75$ (M8)

Braking safety coefficient: $K_Z=1.75$ (M8)

制动器的制动力矩: $M_Z=818\text{N}\cdot\text{m}$

Braking torque of the brake: $M_Z=818\text{N}\cdot\text{m}$

所选制动器的参数如下:

The parameters of the selected brake are as follows:

制动器型号: **YWZ5-400/125**

Brake model: **YWZ5-400/125**

推力器型号: **YTD800-60**

Thruster model: **YTD800-60**

制动器的最大制动力矩: $M_{Z\max}=2000\text{N}\cdot\text{m}$

Maximum braking torque of the brake: $M_{Z\max}=2000\text{N}\cdot\text{m}$

允许制动力矩: $[M_Z]=820\text{N}\cdot\text{m}$

Allowable braking torque: $[M_Z]=820\text{N}\cdot\text{m}$

重量: $m_{31}=167\text{kg}$

Weight: $m_{31}=167\text{kg}$

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个数: $m_z=2$ 台

Qty.: $m_z=2$ sets

制动器选择满足要求!

The brake selection meets the requirements!

1.8.机构起动、制动时间和加速度的验算

1.8 Checking Calculation for Starting Time, Braking Time and Acceleration of Mechanism

1.8.1.满载起动时间和起动平均加速度验算

1.8.1 Checking Calculation for Starting Time and Average Starting Acceleration at Full Load

①起动时间

① Starting Time

$$t_q = \frac{n \sum J}{9.55(M_{dq} - M_{dj})}$$

电动机平均起动转矩倍数: $\lambda_{dq}=1.8$

Average starting torque multiple of the motor: $\lambda_{dq}=1.8$

电动机静阻力矩: $M_{dj}=1014.0\text{N.m}$ $\left(M_{dj} = \frac{P_Q D_0}{2a i \eta}\right)$

Static resistance torque of the motor: $M_{dj}=1014.0\text{N.m}$ $\left(M_{dj} = \frac{P_Q D_0}{2a i \eta}\right)$

电机轴上转动件的转动惯量: $J_2=2.7\text{kg.m}^2$

Rotational inertia of rotating parts on the motor shaft: $J_2=2.7\text{kg.m}^2$

折算到电机轴上的总转动惯量: $\sum J=7.697\text{kg.m}^2$ $\sum J = 1.15(J_1 + J_2) + \frac{P_Q D_0^2}{4ga^2 i^2 \eta}$

Total rotational inertia converted to the motor shaft: $\sum J=7.697\text{kg.m}^2$

$$\sum J = 1.15(J_1 + J_2) + \frac{P_Q D_0^2}{4ga^2 i^2 \eta}$$

满载起动时间: $t_q=1.5$

Starting time at full load: $t_q=1.5$

允许的起动时间: $[t_q]=3.0\text{s}$ 满足要求!

Allowable starting time: $[t_q]=3.0\text{s}$ Meet the requirements!

②起动平均加速度

② Average Starting Acceleration

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平均加速度: $a_q=0.28 \text{ m/s}^2$

Average acceleration: $a_q=0.28 \text{ m/s}^2$

允许的起动平均加速度: $[a_q]=0.30 \text{ m/s}^2$

Allowable average starting acceleration: $[a_q]=0.30 \text{ m/s}^2$

1.8.2.满载制动时间和制动平均加速度验算

1.8.2 Checking Calculation for Braking Time and Average Braking Acceleration at Full Load

①制动时间

① Braking Time

$$t_z = \frac{n' \sum J'}{9.55(m_z M_z - M'_j)}$$

满载下降制动电动机转速: $n'=1654.1 \text{ r/min}$

Rotating speed of descent braking motor at full load: $n'=1654.1 \text{ r/min}$

电动机静阻力矩: $M'_j=861.2 \text{ N.m}$

$$\left(M'_j = \frac{P_0 D_0}{2 a i} \eta \right)$$

Static resistance torque of the motor: $M'_j=861.2 \text{ N.m}$

$$\left(M'_j = \frac{P_0 D_0}{2 a i} \eta \right)$$

折算到电机轴上的总转动惯量: $\sum J' = 7.697 \text{ kg.m}^2$

$$\sum J' = 1.15(J_1 + J_2) + \frac{P_0 D_0^2 \eta}{4 g a^2 i^2}$$

Total rotational inertia converted to the motor shaft: $\sum J' = 7.697 \text{ kg.m}^2$

$$\sum J' = 1.15(J_1 + J_2) + \frac{P_0 D_0^2 \eta}{4 g a^2 i^2}$$

制动时间: $t_z=1.7 \text{ s}$

Braking time: $t_z=1.7 \text{ s}$

允许的制动时间: $[t_z]=3.0 \text{ s}$ 满足要求!

Allowable braking time: $[t_z]=3.0 \text{ s}$ Meet the requirements!

②制动平均加速度

② Average Braking Acceleration

平均加速度: $a_z=0.54 \text{ m/s}^2$

$$\left(a_z = \frac{1.1 v_q}{t_z} \right)$$

Average acceleration: $a_z=0.54 \text{ m/s}^2$

$$\left(a_z = \frac{1.1 v_q}{t_z} \right)$$

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允许的制动平均加速度: $[a_z]=0.55\text{m/s}^2$

Allowable average braking acceleration: $[a_z]=0.55\text{m/s}^2$

1.8.3.空载起动时间和起动平均加速度验算

1.8.3 Checking Calculation for Starting Time and Average Starting Acceleration at No-Load

① 起动时间

① Starting Time

电动机静阻力矩: $M_{dj}=447.9\text{N.m}$ $\left(M_{dj} = \frac{LS \cdot gD_0}{2ai\eta}\right)$

Static resistance torque of the motor: $M_{dj}=447.9\text{N.m}$ $\left(M_{dj} = \frac{LS \cdot gD_0}{2ai\eta}\right)$

折算到电机轴上的总转动惯量: $\Sigma J=7.697\text{kg.m}^2$ $\Sigma J = 1.15(J_1 + J_2) + \frac{LS \cdot D_0^2}{4a^2i^2\eta}$

Total rotational inertia converted to the motor shaft: $\Sigma J=7.697\text{kg.m}^2$

$\Sigma J = 1.15(J_1 + J_2) + \frac{LS \cdot D_0^2}{4a^2i^2\eta}$

满载起动时间: $t_q = 0.9\text{s}$

Starting time at full load: $t_q = 0.9\text{s}$

允许的起动时间: $[t_q] = 3.0\text{s}$ 满足要求!

Allowable starting time: $[t_q]=3.0\text{s}$ Meet the requirements!

② 起动平均加速度

② Average Starting Acceleration

平均加速度: $a_q = 0.28\text{m/s}^2$

Average acceleration: $a_q=0.28\text{m/s}^2$

允许的起动平均加速度: $[a_q] = 0.30\text{m/s}^2$

Allowable average starting acceleration: $[a_q]=0.30\text{m/s}^2$

1.8.4.空载制动时间和制动平均加速度验算

1.8.4 Checking Calculation for Braking Time and Average Braking Acceleration At No-Load

① 制动时间

① Braking Time

满载下降制动电动机转速: $n' = 1654.1 \text{ r/min}$

Rotating speed of descent braking motor at full load: $n'=1654.1 \text{ r/min}$

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电动机静阻力矩: $M'_j = 380.4 \text{ N.m}$ $\left(M'_j = \frac{LS \cdot g D_0}{2ai} \eta \right)$

Static resistance torque of the motor: $M'_j = 380.4 \text{ N.m}$ $\left(M'_j = \frac{LS \cdot g D_0}{2ai} \eta \right)$

折算到电机轴上的总转动惯量: $\Sigma J' = 7.697 \text{ kg.m}^2$

$$\Sigma J' = 1.15(J_1 + J_2) + \frac{LS \cdot D_0^2 \eta}{4a^2 i^2}$$

Total rotational inertia converted to the motor shaft: $\Sigma J' = 7.697$

$$\text{kg.m}^2 \Sigma J' = 1.15(J_1 + J_2) + \frac{LS \cdot D_0^2 \eta}{4a^2 i^2}$$

起动时间: $t_z = 1.1 \text{ s}$

Starting time: $t_z = 1.1 \text{ s}$

允许的起动时间: $[t_z] = 3.0 \text{ s}$ 满足要求!

Allowable starting time: $[t_z] = 3.0 \text{ s}$ Meet the requirements!

②制动平均加速度

② Average Braking Acceleration

$$\text{平均加速度: } a_z = 0.31 \text{ m/s}^2 \left(a_z = \frac{1.1 v_q}{t_z} \right)$$

$$\text{Average acceleration: } a_z = 0.31 \text{ m/s}^2 \left(a_z = \frac{1.1 v_q}{t_z} \right)$$

允许的制动平均加速度: $[a_z] = 0.30 \text{ m/s}^2$ 满足要求!

Allowable average braking acceleration: $[a_z] = 0.30 \text{ m/s}^2$ Meet the requirements!

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